

SCHOOL OF ENGINEERING, CUSAT
First Series Examination April 2024
Semester : IV
MECHANICAL ENGINEERING
19-205-0402: METROLOGY & INSTRUMENTATION

Time: 2 Hours

Max. Marks: 50

Course Outcomes:

1. Understand the concepts of measurements, limits, fits and tolerance.
2. Learn about measurement techniques of angles, tapers, and surface finish
3. Learn about applications of measuring instruments.
4. Understand the measurement techniques for strain, force, torque, temperature, air pollution and nuclear radiation.

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

Part A
Answer ALL questions

4x5= 20 Marks

Q. No.	Questions	Marks	BL	CO	PO
I a)	Differentiate between precision , accuracy and repeatability.	5	L2	1	1,2
b)	Differentiate between tolerance and allowance.	5	L2	1	1,2
c)	Explain the static characteristics of an instrumentation system.	5	L1	3	1
d)	State and explain Taylors principle of limit-gauge design.	5	L1	1	1,2

2X15 = 30 Marks

Part B

Marks BL CO PO

II (a)	Determine the fundamental deviation and tolerances and the limits of size for hole and shaft pair in the fit: 25 mm H8 - d9. The diameter steps are 18 mm and 30 mm. The fundamental deviation for d shaft is given as $-16 D^{0.44}$. The tolerance units is, $i = 0.45 \times \sqrt[3]{D} + 0.001 D$. The tolerance grade for number 8 quality is 25 i and for number 9 quality is 40 i.	15	L3	1	1,2
(b)	What do you mean by interchangeability?	5	L2	1	1,2

OR

(a) Briefly explain different types of fits with neat sketches.

10

L2

1

1,2

(b) Discuss the functional elements of measuring system with a neat block diagram. Draw a liquid-in-glass thermometer and show its functional elements

5

L2

3

1,2

B.Tech Degree IV Semester A Batch
First Internal Examination, April 2024
ME 19-205-0404 APPLIED THERMODYNAMICS
(2019 Scheme)

Time: 2 Hours

Max. Marks: 40

Course Outcomes:

On completion of this course the student will be able to:

1.	Learn first and second law of thermodynamics and the application of these laws for various thermodynamic processes.
2.	Apply the knowledge of thermodynamic laws to various engineering systems.
3.	Analyse the performance of steam generators with additional accessories used for enhancing the performance.
4.	Understand the various types of nozzles and its performance at various back pressure conditions and also for different flow conditions.
5.	Calculate the mole and mass fractions of various mixtures and also various specific properties of the mixtures

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

(Use of Steam Table is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

	Questions	Marks	BL	CO	PO
1.	Prove that the violation of Kelvin Plank statement implies violation of Clausius statement.	5	L2	1	1
2.	Describe various laws for ideal gases - Dalton's law of partial pressure, Amagat's law of partial volumes, Gibbs-Dalton law with respect to enthalpy, entropy and internal energy.	5	L1	5	1
3.	A Carnot heat engine operates between a source temperature of 1000 °C and a sink temperature of 40 °C. Find the least rate of heat rejection per kW net output of the engine.	10	L5	1	1,2,4
4.	A tank of capacity 0.6 m ³ is insulated and divided into two sections through a partition. One section initially contains H ₂ at 3 bar, 130 °C and has a volume of 0.4 m ³ . The other section holds N ₂ at 6 bar and 30 °C. The gases are allowed to mix. Find the final temperature and pressure of the mixture. For Nitrogen, c _v may be taken as 0.744 kJ/kgK and for Hydrogen, c _v may be taken as 10.352 kJ/kgK.	10	L4	5	1,2,4
5.	The following is the gravimetric analysis of the air: Oxygen 23.14%, Nitrogen 75.53%, Argon 1.28% and Carbon dioxide 0.05%. Calculate the volumetric analysis and the partial pressure of each constituent when the total pressure is 1 bar	10	L4	5	1,2,4

A-Batch
GFI

Time: 2hrs	Department of Mechanical Engineering, SOE 1 st Internal Examination of S-4 ME-A Batch - April 2024 19-205-0406 MANUFACTURING PROCESSES	Max Marks:40
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Course Outcomes:

On completion of this course the student will be able to:

1. Learn about the sand casting process.
2. Understand the special casting process and casting defects.
3. Learn about different metal forming processes.
4. Understand different metal joining processes and their characteristics.

Part-A (Answer all questions)

(Max marks= 10)

Q No

- | | CO | BL | Marks |
|--|-----|----|-------|
| 1. Discuss the salient features of a good sand mould. | CO1 | L2 | 2.5 |
| 2. What are the principal constituents of a moulding sand? Briefly explain. | CO1 | L2 | 2.5 |
| 3. Explain the following elements of sand casting:
i. Cope & Drag
ii. Core
iii. Riser
iv. Gate | CO1 | L2 | 2.5 |
| 4. What are the functions of gating system in casting? | CO2 | L2 | 2.5 |

Part-B (Answer all questions)

(Max marks= 30)

- | | | | |
|--|-----|----|---|
| 5. Design a pattern for casting having dimension of length = 200 mm, width = 100 mm and height = 80 mm. It has central circular hole having 80 mm diameter. Casting is made from steel having shrinkage allowance of 20 mm/meters. | CO1 | L3 | 6 |
| 6. Discuss the general procedure for making sand cores. | CO2 | L2 | 6 |
| 7. Explain the steps in investment casting with adequate sketches. | CO2 | L2 | 6 |
| 8. Explain the process, advantages, disadvantages and applications of centrifugal casting with neat figures. | CO2 | L2 | 6 |
| 9. Discuss in detail, the types of cores according to the core material and position of the core print in the mould. | CO1 | L2 | 6 |

**B.Tech Degree IV semester First Internal Examination in Mechanical Engineering,
April 2024**

19-205-0405: HYDRAULIC MACHINERY (2019 Scheme)

Time: 2 Hours

Max. Marks: 88

Course Outcomes:

On completion of this course the student will be able to:

1. Understand dimensional analysis and principles of similitude.
2. Apply the momentum principles to impinging jets and analyse its performance.
3. Learn the working of various hydraulic turbines and their performance characteristics.
4. Understand the working of roto-dynamic pumps and positive displacement pumps.
5. Study the principle of working of hydraulic devices

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

Part A

7x4= 28 marks

Q. No.	Questions	Marks	BL	CO	PO
I a)	What is meant by geometric, kinematic and dynamic similarities?	4	L2	2	1,2
b)	Obtain an expression for the force exerted by a jet of water on the inclined plate moving in the direction of the jet.	4	L2	1	1,2
c)	What do you mean by repeating variables? How are the repeating variables selected for dimensional analysis?	4	L1	1	1
d)	Find the efficiency of a free jet striking normally on a series of flat plates mounted on a periphery of a wheel?	4	L2	2	1,2
e)	Draw a neat sketch and explain the principle and working of a hydraulic press?	4	L1	1	1,2
f)	Obtain the expression for Rankine's efficiency of a hydraulic ram.	4	L1	1	1
h)	State Buckingham's π theorem	4	L1	1	1

Part B

6x10= 60 marks

Q. No.

Questions

Marks BL CO PO

2

What do you mean by dimensionless numbers? Define and explain any two of them.

10 L4 3 1,2,3

Derive on the basis of dimensional analysis suitable parameters to present the thrust developed by a propeller. Assume that the thrust P depends upon the angular velocity m , speed of advance V , diameter D , dynamic viscosity μ , mass density ρ , elasticity of the fluid medium which can be denoted by the speed of sound in the medium C .

3

A jet of water having a velocity of 15 m/s, strikes a curved vane which is moving with a velocity of 5 m/s in the same direction as that of the jet at inlet. The vane is so shaped that the jet is deflected through 135° . The diameter of jet is 100 mm. Assuming the vane to be smooth, find :

10 L4 4 3,4

1. Force exerted by the jet on the vane in the direction of motion,
2. Power exerted on the vane, and
3. Efficiency of the vane.

4

A ship 300 m long moves in sea-water, whose density is 1030 kg/m^3 . A 1:100 model of this ship is to be tested in a wind tunnel. The velocity of air in the wind tunnel around the model is 30 m/s and the resistance of the model is 60 N. Determine the velocity of ship in sea-water and also the resistance of the ship in sea-water. The density of air is given as 1.24 kg/m^3 . Take the kinematic viscosity of the sea-water and air as 0.012 Stokes and 0.018 Stokes respectively.

10 L4 3 3,4

5

Explain with the help of a neat sketch the principle and working of hydraulic coupling and hydraulic torque convertor.

10 L4 5 4,5

6

The pressure intensity of water supplied to an intensifier is 20 N/cm^2 while the pressure intensity of water leaving the intensifier is 100 N/cm^2 . The external diameter of the sliding cylinder is 20 cm. Find the diameter of the fixed ram of the intensifier

10 L4 5 4

7

Explain with the help of neat sketch, the principle and working of the following hydraulic devices:

10 L3 5 5

(i) Hydraulic lift

(ii) Hydraulic crane

SCHOOL OF ENGINEERING, CUSAT

First Series Examination April 2024

Semester : IV

MECHANICAL ENGINEERING

19-200-0401: COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

Time: 2 Hours

Max. Marks: 50

Course Outcomes:

1. Transform a region to another region using conformal mapping
2. Evaluate real integrals using residue theorem
3. Form and solve partial differential equation
4. Determine solution of partial differential equation for vibrating string and heat conduction

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

Part A

4x5= 20 Marks

Answer ALL questions

Q. No.	Questions	Marks	BL	CO	PO
I	a) If $f(z)$ is an analytic function, then prove that $f'(z)$ is independent of $\bar{z}$.	5	L2	1	1
	b) Find the domain of analyticity of the function $f(z) = \frac{z}{ z ^2}$.	5	L2	1	1
	c) Evaluate $\int_C f(z) dz$ where $f(z) = \frac{z^2}{z^2-1}$ and C is the circle $ z+i =1.4$.	5	L3	2	2
	d) State Cauchy's Integral Formula and evaluate $\int_C \frac{\cos 2z}{4z} dz$ where C is the circle $ z =1$.	5	L3	2	2

2x15 = 30 Marks

Part B

Marks BL CO PO

II	a) Prove that if the function $f(z) = u(x,y) + iv(x,y)$ is analytic within a domain D , then both u and v satisfy Laplace's equation.	7	L2	1	1
	b) Find a bilinear transformation which maps the points $z_1 = -2$, $z_2 = 0$, $z_3 = 2$ into the points $w_1 = 0$, $w_2 = i$, $w_3 = -i$.	4	L3	1	2
	c) Find the image of the line $y - x + 1 = 0$ under the mapping $w = 1/z$.	4	L3	1	2

OR

III	(a) Find an analytic function $f(z) = u + iv$ whose real part is $e^{-x}[x \cos y + y \sin y]$.	7	L3	1	2
	(b) Show that under the transformation $w = z + \frac{1}{z}$ circles with a radius $r = 1$ are transformed onto the segment $-2 \leq u \leq 2$ of the u -axis, while circles with radius $r \neq 1$ are transformed into ellipses. Also, find the points at which the mapping is not conformal.	8	L3	1	1

IV

- (a) Verify Cauchy's integral theorem for the integral of z^2 over the boundary of the square with vertices $1 + i, -1 + i, -1 - i, 1 - i$ in the counterclockwise sense.

10 L4 2 1,2

- (b) Evaluate $\int_C \frac{e^z}{z-2} dz$ where C is $|z| = 3$.

5 L5 2 2

OR

V

- (a) Using Cauchy's integral formula, evaluate $\int_C \frac{e^z}{(z-1)(z-4)} dz$, where C is the circle $|z| = 2$.

7 L3 2 1

- (b) Evaluate $\int_C \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz$ where C is the circle $|z| = 2$.

8 L3 2 1

Answer All Questions

PART A

No	Question	CO	BL	Marks
1. a	Describe the working of thermistor and RTDs. What are their relative merits and demerits	CO1	L1	3
b	Describe the working of an incremental encoder. Explain how the encoder detects direction of rotation	CO1	L2	3
c	Explain the working of a vane pump	CO1	L2	3
d	Find the transfer function, $G(s) = \frac{C(s)}{R(s)}$, corresponding to the differential equation $\frac{d^3c}{dt^3} + 3\frac{d^2c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2r}{dt^2} + 4\frac{dr}{dt} + 3r$. $c(t)$ is the output and $r(t)$ is the input	CO2	L3	3
e	Draw the block diagram representation of the water level control system shown	CO2	L3	3

PART B

No	Question	CO	BL	Marks
2	What will be the change in resistance of an electrical resistance strain gauge with a gauge factor of 2.1 and resistance 50Ω if it is subjected to a strain of 0.001?	CO1	L3	5
3	A single acting pneumatic cylinder positioned vertically is used to lift weights. It is controlled by a three-port two-position direction control valve which has a solenoid on one side and a spring on the other. The cylinder should raise the weight when the solenoid is charged; and when the solenoid is off, the weight is lowered due to gravity. Draw the hydraulic circuit diagram for the case.	CO1	L3	7.5
4	Consider a system represented by the differential equation $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = 2r(t)$; where the initial conditions are $y(0) = 0$, and $\frac{dy}{dt}(0) = 1$. The input is a unit ramp $r(t) = t$, for $t \geq 0$. What is the output $y(t)$?	CO2	L4	7.5
5	Determine the Transfer Function $G(s) = \frac{I_2(s)}{V(s)}$ of the electrical circuit shown. Assume $R_1 = 5\Omega$, $R_2 = 10\Omega$, $L = 2H$, $C = 0.5F$	CO2	L4	7.5
6	The transfer function of a system is given as $G(s) = \frac{Y(s)}{X(s)} = \frac{s}{(s+4)(s+8)}$. What will be the response $y(t)$ if the input to the system is a unit step, i.e., $x(t) = 1$? What is the steady state value of the output?	CO3	L4	7.5

